Chapter 3: Learning from Data

Supervised learning uses labeled examples to learn a mapping from features to labels.

Regression predicts continuous values such as prices; classification predicts discrete categories such as spam or not spam.

A dataset is divided into training, validation and test sets. The validation set is used to tune hyperparameters, and the test set is used once at the end to estimate generalization error.

3.2 Overfitting and Regularization

Overfitting occurs when a model fits noise in the training set, giving low training error but high test error. Underfitting occurs when the model is too simple, giving high error on both.

Regularization adds a penalty on the size of the weights to the loss. L2 regularization (ridge) penalizes the sum of squared weights, shrinking them smoothly. L1 regularization (lasso) penalizes absolute values and can drive weights to exactly zero.

Other remedies include early stopping, dropout for neural networks, and gathering more training data.